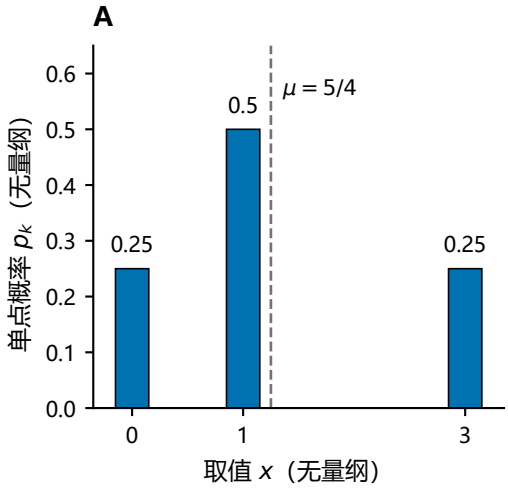
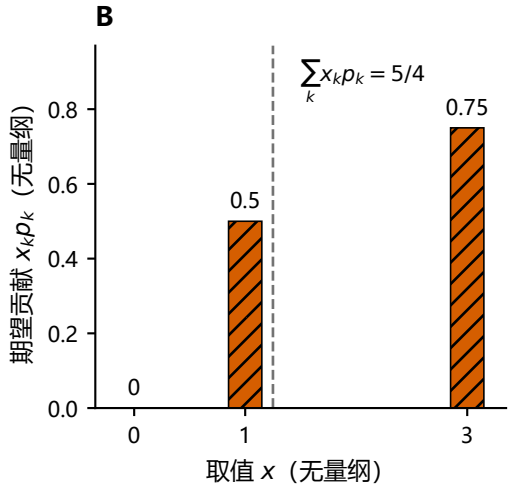


$$\sum_k p_k = 1$$



$$\mathbb{E}[X] = 5/4, \quad \mathrm{Var}(X) = 19/16$$